ARC on DeBERTa-v3

with various retrieval methods

By Andy Zhu

AI2 Reasoning Challenge (ARC)

"Think You Have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge" (arXiv 1803.05457) (Clark et al. 2018)

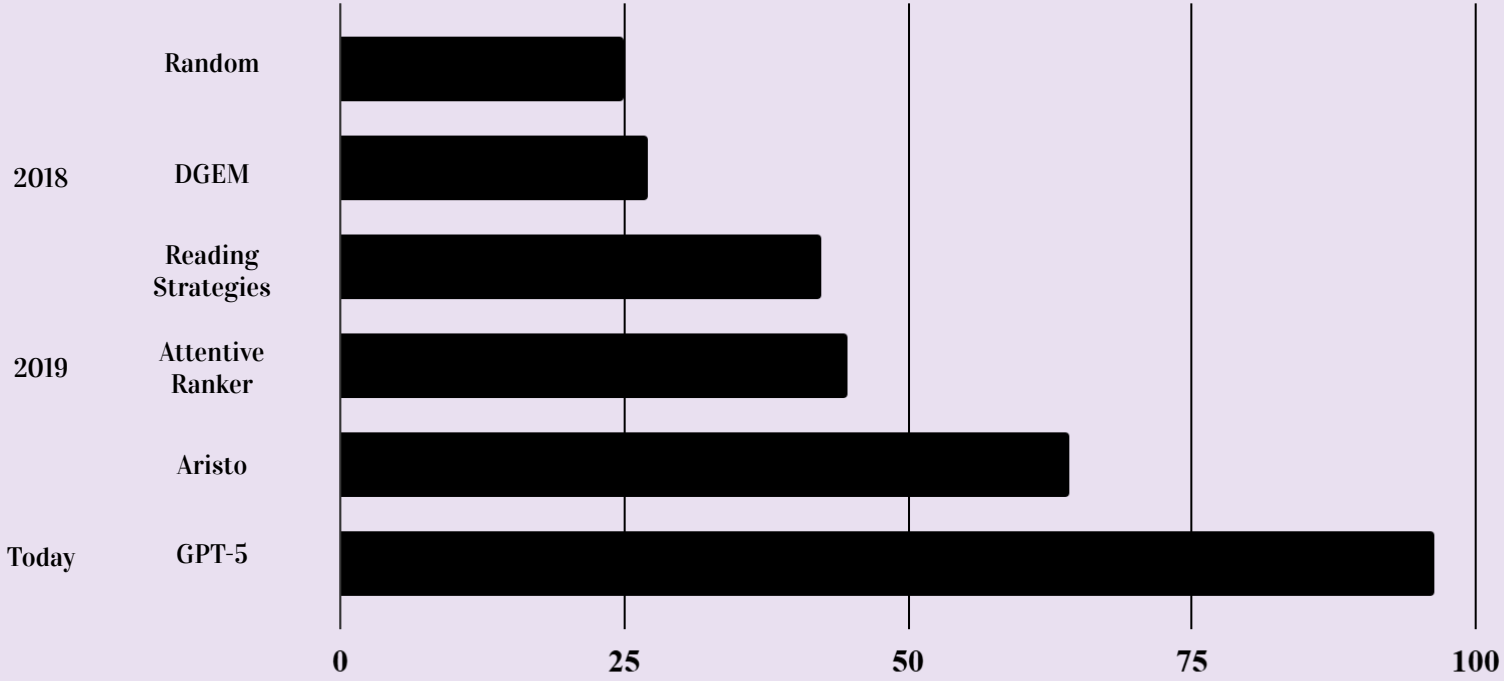
7,787 **Questions**
grade-school level science questions from real standardized exams

2 **Splits**
“Easy” and “Challenge” determined by simple IR and PMI baselines

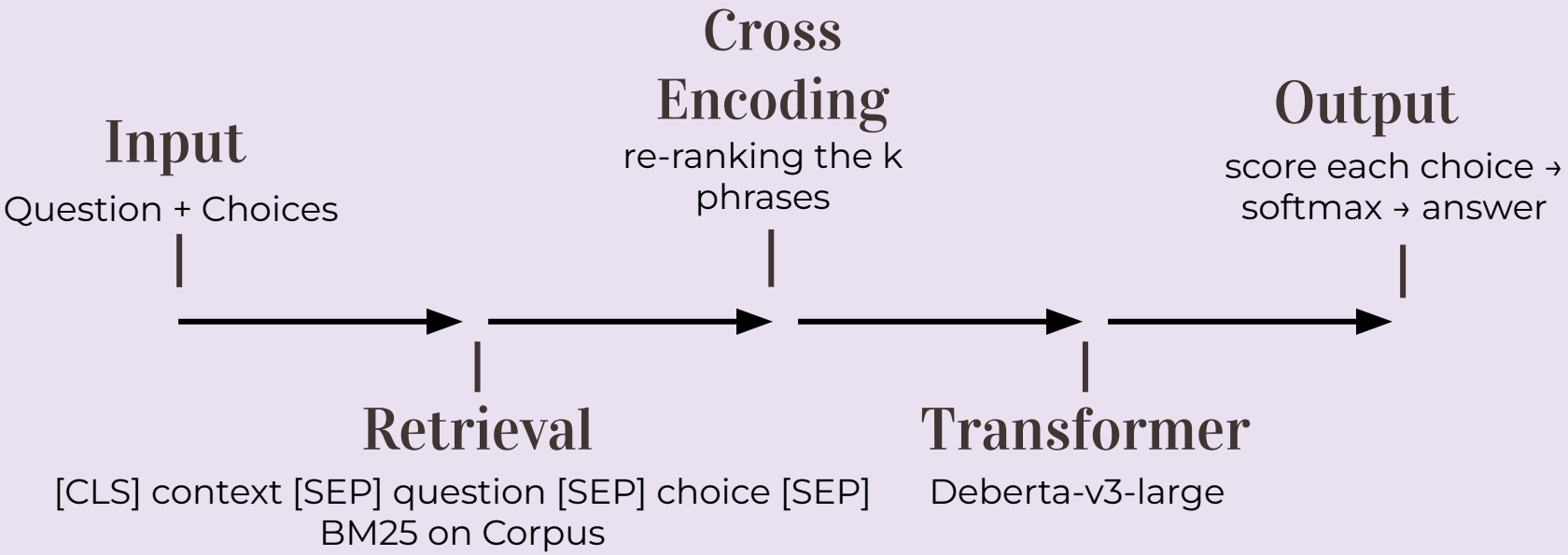
14M **Corpus**
Science Questions scraped from the web

25% **Average Score**
on most models back in the day

Accuracies over the years



Method Pipeline



Other Methodologies

Warmup

Prevents early large gradients

Ensembling

Average the softmax probability from multiple runs

Combined Training

Combine Easy and Challenge for training Challenge

Choice Masking

Pad answers to 5 slots, mask out the pads.

Pairing

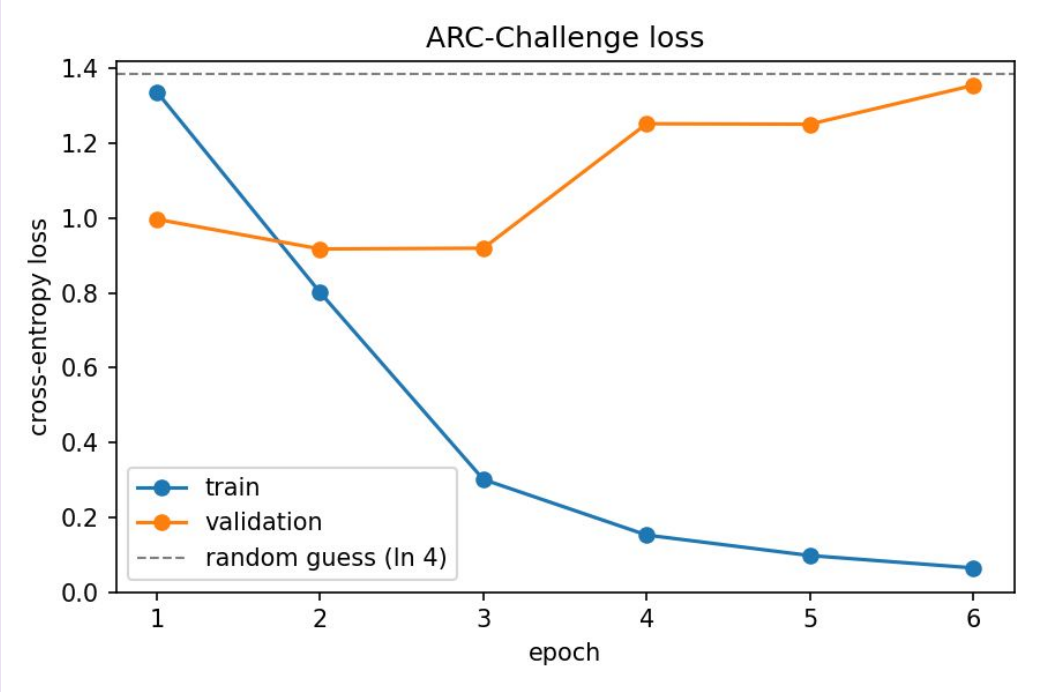
Each question-option pair gets one query

Truncation

Truncate contexts, not questions or choices

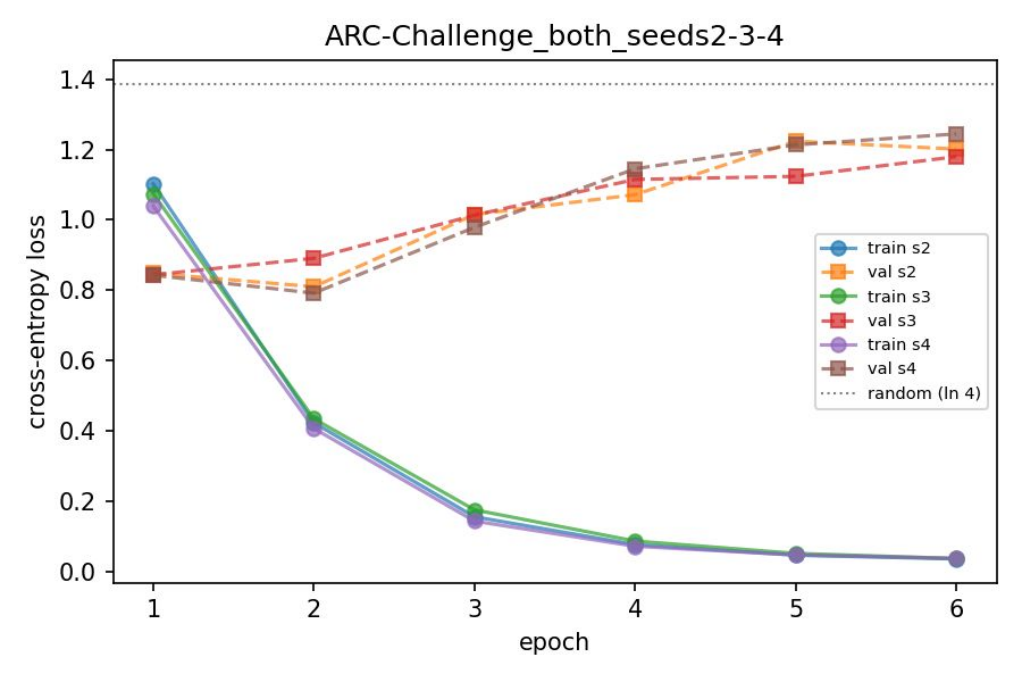
Results / Plots

0.684



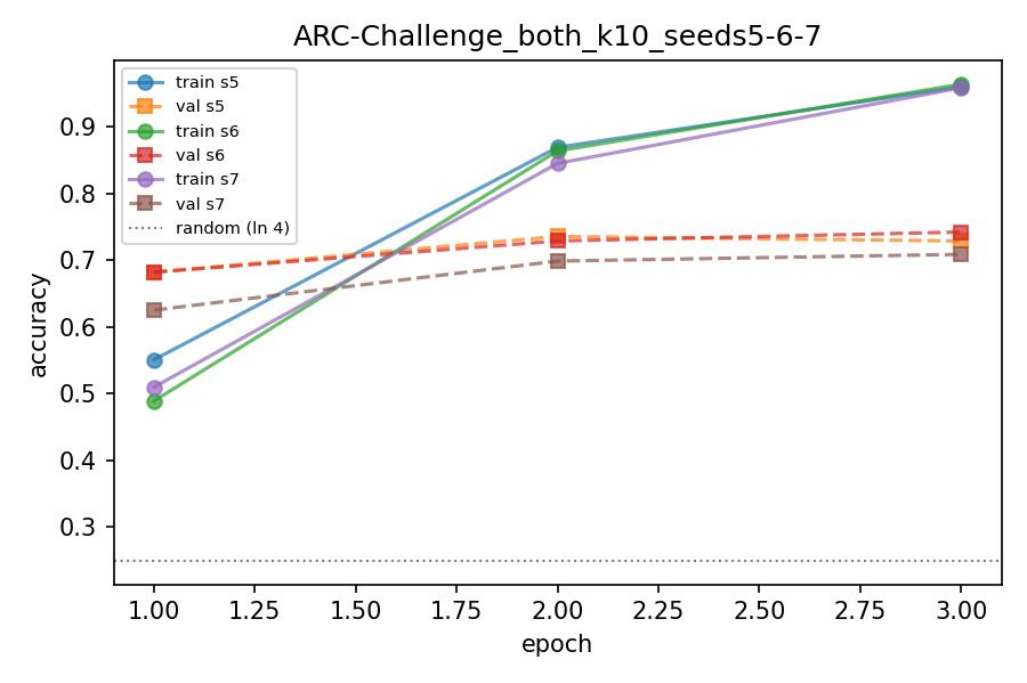
Results / Plots

0.7244 ± 0.0023 | 0.737

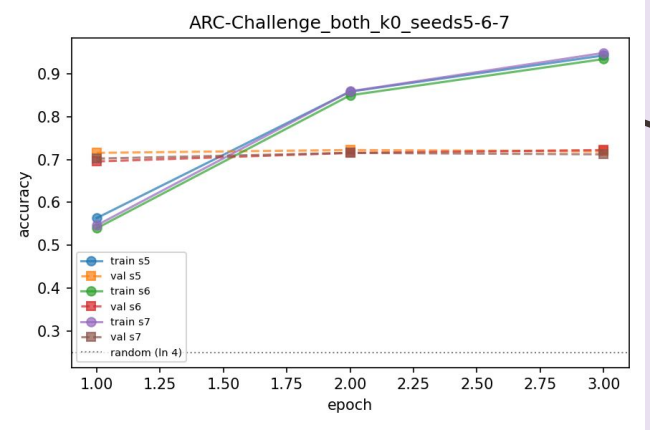


Results / Plots

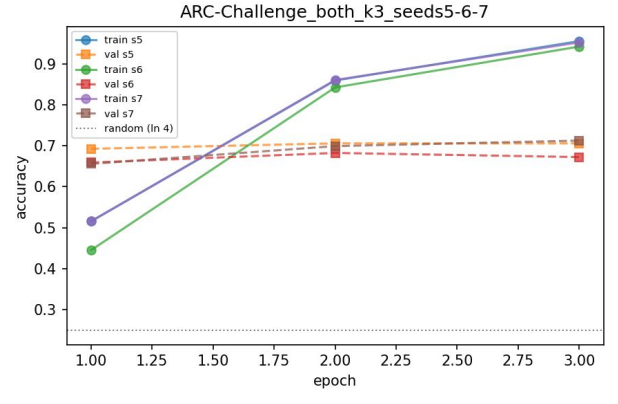
0.7226 ± 0.0123 | 0.7440



0.7150 ± 0.0095 | 0.737

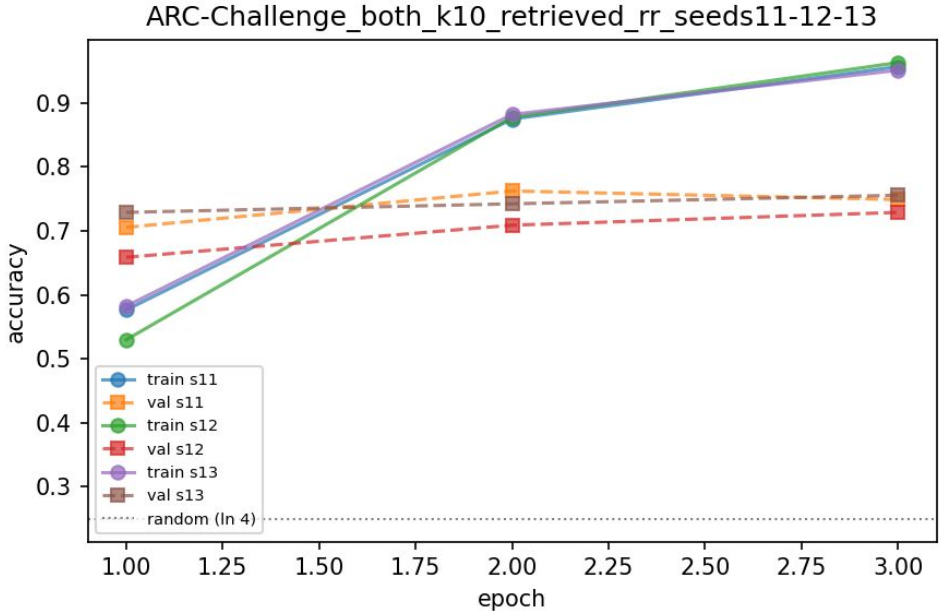


0.7119 ± 0.0110 | 0.7329

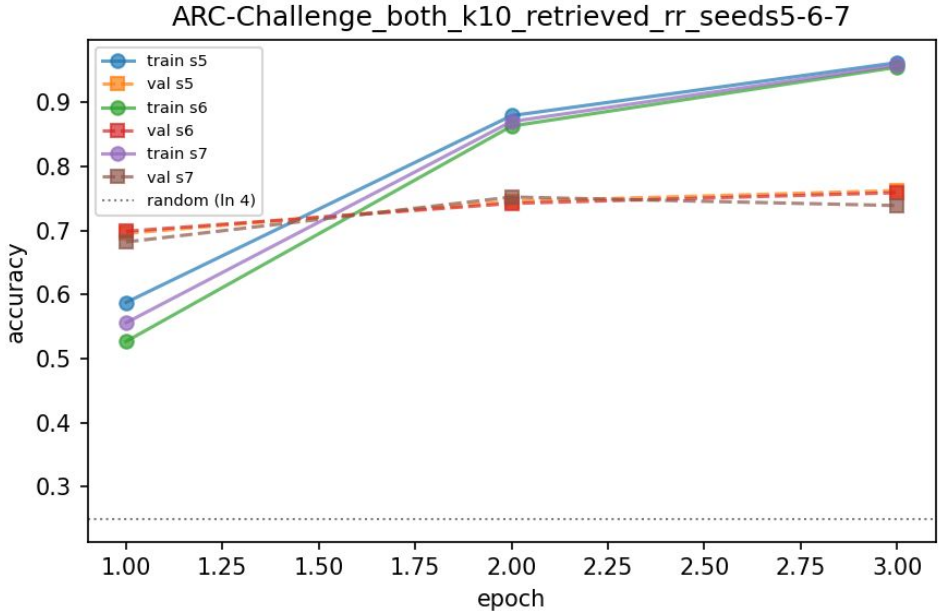


Results / Plots

0.7398 ± 0.0068 | 0.7509



0.7400 ± 0.0038 | 0.7577



Conclusion

- DeBERTa-v3-large did most of the work, ~70% baseline accuracy
- Retrieval doesn’t help that much
- Re-ranking did help, it also dropped the variance.

Thats all